
A Century Is Not Enough: Certification Bounds for Out-of-Sample Return Predictability

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Abstract

How much predictability could a century of data rule out? We answer in closed form and then measure it. Evidence about an alpha accumulates as log wealth, which grows at half the squared information ratio, so a sample spanning T years can certify an annualised information ratio no smaller than a floor that depends on T , the level, and the size of the search—and on nothing else. Sampling the same span more finely does not lower it: the floor for the Goyal–Welch data is 0.38 monthly, 0.37 quarterly and 0.35 annually, across a twelvefold range in observation count. Against the effect Campbell and Thompson call economically significant, an annualised ratio of 0.25, a single predictor needs 99 years to reach 80% power and the 46-predictor search of Goyal et al. [1] needs 244. The canonical dataset is 100 years old. We then apply an anytime-valid certificate to 117 predictor–frequency pairs spanning 1926–2025. Nothing is certified, under any of nine specifications; the largest evidence value ever observed is 8.0 against the 20 required. The deliverable is not that null but the bound beside it: the median predictor’s incremental annualised information ratio is at most 0.60, and 112 of 116 ceilings lie above 0.25, so for almost every published predictor a century of the best-studied data in finance can separate neither “nothing” nor “economically meaningful” from the other. Because evidence compounds, the wealth path also dates each edge: the valuation ratios d/p , d/y and e/p —the predictors a Clark–West test ranks highest—stopped contributing in the 1950s and have added nothing since, which a static test cannot see. Our pipeline reproduces Goyal et al.’s published out-of-sample R_{OS}^2 with a correlation of 0.968 across their fourteen classic predictors.

1 Introduction

The out-of-sample predictability literature asks whether a variable forecasts the equity premium. After Welch and Goyal [2] the consensus answer is mostly no, and after Goyal et al. [1] it is mostly no again for a further twenty-nine candidates. This paper asks a prior question that determines what any such answer can mean: *given a sample of a certain length, what size of effect could it have detected at all?*

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The question has an exact answer, and the answer is uncomfortable. Section 3 derives a certification floor: the smallest annualised information ratio a sample spanning T years can establish at level α . Two features of it matter. The floor scales with $1/\sqrt{T}$ in *calendar span* and is almost unaffected by sampling frequency, so a century of monthly data and a century of annual data face nearly the same bound despite a twelvefold difference in observations. And it degrades with the size of the search, which in this literature is large and known.

Put those together against the effect the literature itself treats as meaningful—Campbell and Thompson [3] take a monthly out-of-sample R_{OS}^2 of 0.5%, an annualised information ratio of about 0.25, as economically significant—and the arithmetic is stark. A single predictor needs 99 years to reach 80% power against it. Correcting for the 46-predictor search that Goyal et al. [1] actually ran needs 244. The canonical dataset begins in 1926.

What we do with that. A bound on what a study could have found is only useful if paired with a bound on what it did find. Section 4 uses an anytime-valid *certificate* [4], a non-negative wealth process whose validity is a finite-sample theorem and whose logarithm is the profit and loss of an explicit drift-hedged strategy. Two properties earn its place here. It supplies a time-uniform *ceiling* on a predictor’s incremental value, which is the number this literature has never reported and the one a reader can act on. And because evidence compounds rather than averages, its wealth path *dates* an edge: a predictor that worked until 1957 and not since cannot keep accumulating, and the path shows exactly when it stopped.

What we find. Across 117 predictor–frequency pairs from 1926 to 2025, nothing is certified, under any of nine specifications of burn-in, window and payoff. That is unsurprising and is not the contribution. The contribution is the interval: the median predictor’s incremental annualised information ratio is bounded above by 0.60, and 112 of 116 ceilings sit above Campbell and Thompson’s 0.25. A century of the most heavily studied data in finance can rule out neither zero nor an economically meaningful edge, for almost every predictor in the set.

The dating result is the one we did not expect. Clark–West and the certificate rank the predictors close to inversely, and the wealth paths say why. The valuation ratios d/p , d/y and e/p —the most cited predictors in the literature, and the ones Clark–West scores highest at 1.34 to 1.64—peaked in 1946–1957 and have contributed nothing in the seventy years since. Inflation and the Treasury-bill rate, which Clark–West scores lowest, are the only variables still accumulating. A test that averages a covariance over a century cannot distinguish a steady edge from a dead one; a test that compounds cannot avoid it.

What this is not. We do not claim the certificate is more powerful than Clark–West. It is not: Guven [4] measures the exchange rate at roughly a factor of two in sample size, the price of validity at every stopping time and under weaker assumptions. Nor do we claim to overturn Welch and Goyal [2]; we agree with it and reproduce it. The claim is narrower and, we think, more useful: this literature has been operating at or below the detection threshold of its own data for its entire history, the threshold is computable in advance, and the quantity worth reporting is a ceiling rather than a p -value.

2 Related work

The predictability literature and its instability. Welch and Goyal [2] evaluate seventeen predictors of the equity premium and find that essentially none beats the historical mean out of sample; Goyal et al. [1] repeat the exercise for twenty-nine further candidates published since, and reach the same verdict for most of them. Campbell and Thompson [3] argue that weak predictability can still be economically meaningful once forecasts are constrained by economic priors, and set the benchmark effect size we use throughout. Rapach et al. [5] recover out-of-sample gains from forecast combination. Running underneath all of it is instability: the same predictor works in one era and not the next, which is the observation our dating result in Section 6 makes precise and Proposition 3 prices.

Power, and what a test can be expected to find. Inoue and Kilian [6] show that out-of-sample tests are typically *less* powerful than in-sample ones, because splitting a finite sample costs infor-

mation, and warn against reading out-of-sample failure as evidence that in-sample evidence was spurious. Our contribution is complementary and more absolute: their comparison is between two procedures on the same data, whereas Proposition 1 bounds what *any* procedure can establish from a span of a given length. Hansen et al. [7] address a related selection problem by constructing a set of models that cannot be distinguished from the best. Where a model confidence set answers “which of these can be told apart”, we answer “could any of them have been told apart from nothing”.

Nested forecast comparison. Diebold and Mariano [8] is not valid for nested models; Clark and McCracken [9] and Clark and West [10, 11] supply the corrections this literature uses, and Guven [4] shows that the benchmark against which the correction is applied determines what null is being tested. We use Clark–West throughout as the incumbent and do not propose to replace it.

Game-theoretic statistics. The certificate is built from test martingales and e-values [12–15], with merging rules from Vovk and Wang [16] and false-discovery control from Wang and Ramdas [17]. Applications of betting arguments to finance exist—Kumon et al. [18] test whether observed *price processes* are martingales using limit-order betting strategies—but that is a specification test of the price process rather than a comparison of a forecasting model against an estimated benchmark, and it does not treat the drift as a nuisance. We are not aware of prior anytime-valid analysis of equity-premium predictability.

3 The certification bound

Let a signal-weighted strategy have annualised information ratio IR, and consider testing whether $IR > 0$ from T years of data at level α .

Proposition 1 (Certification floor). *A one-sided fixed-sample test at level α attains power β against an alternative IR if and only if $T \geq (z_\alpha + z_\beta)^2 / IR^2$ years. Equivalently, a sample spanning T years cannot establish any ratio below*

$$IR_{\min}(T, \alpha, \beta) = \frac{z_\alpha + z_\beta}{\sqrt{T}}.$$

Under a Bonferroni correction for a search over m candidates, replace α by α/m .

The proposition is elementary and that is the point: it needs no new theory and it applies to the tools this literature already uses. Its content is in the substitutions.

Corollary 2 (Frequency does not help). *$IR_{\min}$ depends on the calendar span T and not on the sampling frequency. Increasing the number of observations by sampling a fixed span more finely leaves the floor unchanged.*

This is worth stating because the opposite is widely assumed. Doubling the observation count by moving from quarterly to monthly data halves the standard error of a *per-period* mean, but the annualised information ratio is a per-period quantity scaled by $\sqrt{P}$, and the two effects cancel exactly. Section 6 confirms it on the data: the measured floor is 0.38, 0.37 and 0.35 at monthly, quarterly and annual frequency over the same century.

Proposition 1 assumes the effect is present throughout. It is not, in this literature, and the correction is unfavourable.

Proposition 3 (Intermittent effects raise the floor). *Suppose the conditional mean is present only in a fraction ϕ of periods—a duty cycle—with per-period information ratio IR_{on} when present and zero otherwise. A strategy sized by the signal then presents an unconditional ratio*

$$IR_{\text{measured}} = \sqrt{\phi} IR_{\text{on}},$$

so the span required to certify it scales as $1/(\phi IR_{\text{on}}^2)$: a factor $1/\phi$ longer than for an always-present effect of the same strength.

Proof. With signal $g_t = m_t$ and $m_t = m\mathbf{1}\{\text{on}\}$, the per-period expected profit is $\mathbb{E}[g_t y_t] = \phi m^2$ and its variance is $\mathbb{E}[g_t^2] \sigma^2 = \phi m^2 \sigma^2$ to leading order, so the ratio is $\phi m^2 / \sqrt{\phi m^2 \sigma^2} = \sqrt{\phi} (m/\sigma)$. Substituting into Proposition 1 gives the span. Simulation confirms the identity to within three per cent at $\phi \in \{1, 0.5, 0.25, 0.14\}$. $\square$

Table 1: Years of data needed to reach 80% power against the effect Campbell and Thompson [3] call economically significant—a monthly out-of-sample R_{OS}^2 of 0.5%, an annualised information ratio of 0.25—as the search widens. The searches are those the literature actually ran. The Goyal–Welch sample begins in 1926.

Correction	Floor at 100 years	Years needed for IR = 0.25
One predictor, $\alpha = 0.05$	0.252	99
Bonferroni over 17 [2]	0.365	207
Bonferroni over 46 [1]	0.397	244
Bonferroni over 100	0.420	273
<i>Available: 100 years (1926–2025).</i>		

Table 2: The same effect in four currencies. Monthly out-of-sample R_{OS}^2 as reported by the literature; the implied per-period and annualised information ratios; and the calendar span a one-sided fixed-sample test needs to reach 80% power against it, uncorrected and Bonferroni-corrected for the 46-predictor search of Goyal et al. [1].

Monthly R_{OS}^2	Per-period IR	Annualised IR	Years, one test	Years, 46 tests
0.1%	0.032	0.11	515	1271
0.2%	0.045	0.16	257	635
0.5%	0.071	0.25	103	253
1.0%	0.101	0.35	51	126
2.0%	0.143	0.49	25	62

This is where the two halves of the paper meet. Section 6 shows that the valuation ratios stopped contributing after the 1950s, which on a 1946–2025 evaluation window is a duty cycle of roughly 0.14. An on-period annualised ratio of 0.5—a strong effect by this literature’s standards—would then present unconditionally as 0.19 and require about 178 years to certify. For e/p , whose wealth path is flat from 1946 onwards, the implied requirement is in the thousands of years. Intermittency is not a footnote to the sample-size problem; it is a multiplier on it.

Table 1 and the left panel of Figure 1 state the position. A single predictor tested once sits within one year of the boundary, with no margin for parameter instability, transaction costs, or a slightly smaller true effect. Corrected for the search that was actually performed, the literature is below the line by a factor of two to three in required span.

Reading the floor in the units the literature uses. Table 2 converts a monthly out-of-sample R_{OS}^2 —the quantity this literature reports—into the annualised information ratio the bound is stated in, and then into required span. The row that matters is the third: Campbell and Thompson’s threshold for economic significance corresponds to an annualised ratio of 0.25 and needs 103 years for a single predictor, 253 once the search is corrected for. Even a monthly R_{OS}^2 of 1%, twice what that literature treats as meaningful and larger than anything in our results, needs 126 years under the correction.

4 The instrument

We use the drift-robust certificate of Guven [4] and summarise only what is needed here. For a predictable signal g_t —the model’s forecast minus the intercept-only forecast from the same estimation window—and an outcome y_t , the certificate is

$$\mathcal{E}_t = \inf_{\mu \in \mathbb{R}} \left[w W_t^{\text{sig}}(\mu) + (1 - w) W_t^{\text{ctr}}(\mu) \right],$$

where W^{sig} is the wealth of a bettor staking a predictable fraction on g_t leading y_t , and W^{ctr} is the wealth of a bettor staking two-sidedly on μ being the wrong drift. Each is a non-negative martingale for fixed μ , so their convex combination is one, and an infimum is at most its value at the truth; Ville’s inequality then gives $P(\exists t : \mathcal{E}_t \geq 1/\alpha) \leq \alpha$ in finite samples and uniformly in time, for any drift. We use the identity payoff throughout, which assumes only a martingale-difference null and an a-priori envelope on $|y_t|$, and report the bounded payoffs as sensitivities.

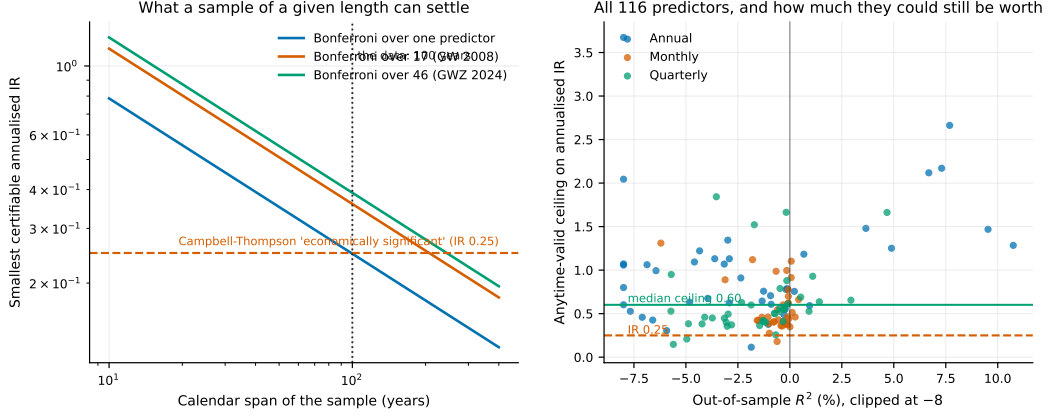


Figure 1: **Left:** the certification floor against calendar span, for searches of one, seventeen and forty-six predictors. The horizontal line is Campbell and Thompson’s economic-significance threshold; the vertical line is the data. **Right:** every predictor–frequency pair, plotted as its anytime-valid ceiling against its out-of-sample R^2_{OS} . The median ceiling is 0.60 and 112 of 116 lie above 0.25; for almost every predictor the data rule out neither zero nor an economically meaningful edge.

Three properties are load-bearing here.

A ceiling, not just a verdict. Inverting the same construction over the value parameter gives a time-uniform confidence interval for $\theta = \mathbb{E}[z_t(y_t - \mu)]$, the expected return of a position sized by the standardised signal. Read as an annualised information ratio, its upper endpoint is the number we report.

Evidence compounds, so it can be dated. $\mathcal{E}_t$ is a product. A predictor whose edge stops contributing stops the process, and the path records when.

Multiplicity without a bootstrap. E-values average under arbitrary dependence, so the grid-level statistic over 117 dependent tests is an average, and false-discovery control follows from e-BH [17].

5 Data and design

We use the Goyal et al. [1] dataset through December 2025: monthly, quarterly and annual equity-premium series from 1926 together with the seventeen classic Welch and Goyal [2] predictors and the twenty-nine added by Goyal et al.. The outcome is the log equity premium $\log(1 + r_t) - \log(1 + r_t^f)$; predictors are lagged one period; forecasts come from an expanding-window OLS regression with an intercept, re-estimated every period, and the benchmark is the same window’s historical mean, which is that regression’s intercept-only restriction. Following Goyal et al., the out-of-sample period begins twenty years after the in-sample period starts.

Real-time versus revised series. Many of the new predictors require re-estimation every period, and Goyal et al. ship them as vintage matrices whose rows index the period and whose columns index the vintage. The admissible series is the diagonal—what a forecaster actually had—and the revised series is the last column. We use the diagonal throughout and record the source per predictor. On the twenty-five predictors where both exist, the revised series looks *better* for only four, so this choice does not drive the results; it is nonetheless the difference between an out-of-sample claim and a claim that quietly uses full-sample parameter estimates.

Validation. Because everything here depends on the pipeline being right, we validate it against Goyal et al.’s published Table 3 Panel A, predictor by predictor, matching their evaluation window and their Campbell–Thompson restriction. Across the fourteen classic monthly predictors the correlation is 0.968 (Spearman 0.959), signs agree in 13 of 14, and the median absolute gap is 0.075 percentage points. The single disagreement is *tms*, where both values round to zero.

Table 3: Specification sensitivity, fourteen classic monthly predictors. Rejecting needs an e-value of 20.

Burn-in	Window	Payoff	Median R_{OS}^2 (%)	Max e-value	Grid e-value	Certified
10 y	expanding	identity	−0.329	3.05	1.48	0
20 y	expanding	identity	−0.378	5.71	2.04	0
30 y	expanding	identity	−0.342	8.02	2.55	0
40 y	expanding	identity	−0.308	6.27	1.94	0
20 y	rolling 20 y	identity	−0.989	5.50	1.81	0
20 y	rolling 30 y	identity	−0.579	8.04	2.73	0
20 y	rolling 50 y	identity	−0.515	7.58	2.17	0
20 y	expanding	tanh	−0.378	1.74	1.24	0
20 y	expanding	sign	−0.378	1.38	1.08	0

Remark 4 (The Campbell–Thompson restriction is not free). Truncating the forecast at a non-negative equity premium helps 13 of the 14 predictors and moves the mean out-of-sample R_{OS}^2 from −0.511% to −0.281%. Since the effects under discussion are themselves a few tenths of a percentage point, an economically motivated truncation is doing about as much work as any predictor in the set. It helps most exactly where the unrestricted regression produces the wildest forecast. We report unrestricted numbers by default.

6 Results

Nothing is certified, and not for want of trying. Across 117 predictor–frequency pairs, out-of-sample R_{OS}^2 is negative for 95 (81%), no predictor reaches the $\alpha = 0.05$ threshold, the grid-level e-value is 1.79 against the 20 required, and nothing survives e-BH. Table 3 sweeps the specification: burn-ins of ten to forty years, rolling windows of twenty to fifty years against the expanding one, and two alternative payoffs. Nothing certifies under any of the nine, and the largest e-value ever observed is 8.04. The rolling windows are the informative negative—Welch and Goyal’s complaint is instability, so a window that forgets is the specification most likely to find something, and it does not.

The ceilings are the result. The right panel of Figure 1 plots every pair. The median ceiling on the incremental annualised information ratio is 0.60; 112 of 116 exceed 0.25. By frequency the medians are 0.46 monthly, 0.53 quarterly and 0.86 annual. The correct summary of a century of this literature is therefore not “the predictors do not work” but a statement with a number in it: *the incremental information ratio of the median canonical predictor lies somewhere between zero and approximately 0.6, and a hundred years cannot narrow it further.*

Frequency does not buy power. The certifiable floor is 0.38 monthly, 0.37 quarterly and 0.35 annually over the same span, across 1200, 400 and 100 observations respectively—Corollary 2, measured.

When each edge existed. Table 4 and Figure 2 give the result we did not anticipate. Clark–West averages a covariance over the century; the certificate compounds, so a dead edge stops the process and the path dates the death. The three valuation ratios have added nothing since the 1950s. The two variables still accumulating are the two Clark–West likes least.

6.1 Could the samples ever have supported the claims?

The bound is a statement about a sample, so it can be applied retrospectively. Every variable in the set has a recorded data span, that span fixes the floor any test faces on it, and the comparison with a claimed effect is then arithmetic rather than opinion.

Table 5 does this for the 56 variables with a recorded sample. Of those, 23—41%—have a span too short to certify an annualised ratio of 0.25 even as a single test at the 5% level. Once the 46-variable search that generated the set is corrected for, the figure is 56 of 56. The median span is 99 years and the median corrected floor is 0.39, against a target of 0.25.

Table 4: Clark–West against the certificate, and what the wealth path shows. “Nats” are the log-wealth contributed in each half of the evaluation period.

Predictor	Clark–West	e-value	Peak year	Nats, 1st half	Nats, 2nd half
<i>d/y</i>	1.64	1.14	1957	+0.13	+0.00
<i>e/p</i>	1.55	1.00	1946	+0.00	+0.00
<i>tbl</i>	1.44	3.48	2021	+0.20	+1.05
<i>d/p</i>	1.34	1.45	1957	+0.37	+0.00
<i>tms</i>	0.78	3.92	1987	+1.34	+0.03
<i>infl</i>	0.83	5.71	1999	+0.59	+1.16

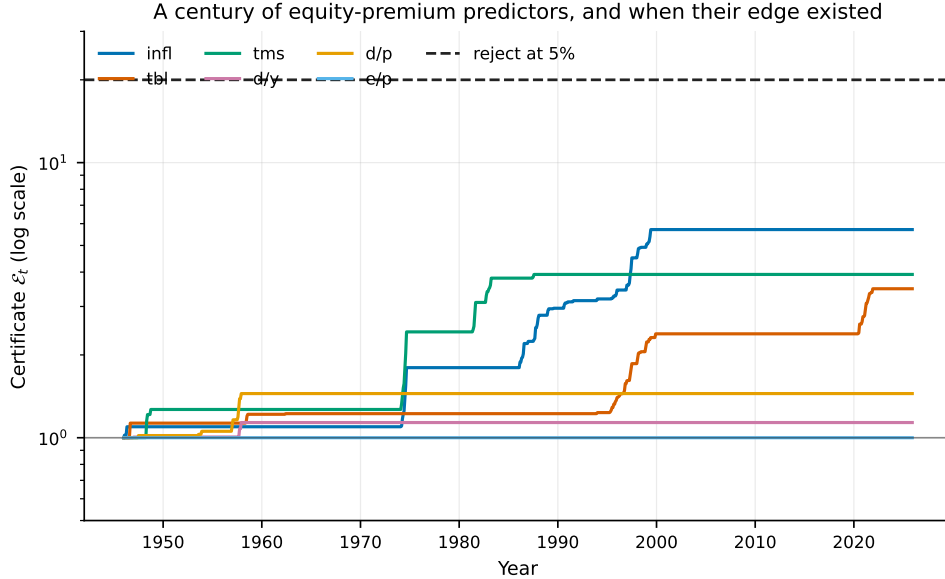


Figure 2: Certificate wealth paths for six classic predictors. A predictor whose edge stops contributing stops compounding, so the path dates it. The valuation ratios flatten in the 1950s; inflation and the Treasury-bill rate are the only two still rising.

The direction of the shortfall is the unfavourable one. The shortest samples belong to the newest variables: the risk-neutral variance measures span under thirty years and face a solo floor of 0.46, nearly twice the effect they would need to establish. A literature that adds variables faster than it adds data moves away from the threshold, not towards it.

6.2 How much of the time was each edge present?

Proposition 3 needs a measured duty cycle to bite. We take it from the certificate’s underlying process—not from $\mathcal{E}_t$, which is a running maximum and so rises only when a new high is set—by splitting each evaluation window into ten equal sub-periods and recording the fraction in which log wealth increased. A predictor with no edge scores about one half.

The median across 116 predictors is 0.40, with quartiles 0.30 and 0.50: slightly *below* the no-edge baseline, which is what a set of variables with no reliable edge looks like. More telling, 92 of 116—79%—reach their peak wealth before the final quarter of their own evaluation window, meaning they contributed nothing over their last quarter-century or more.

Feeding the median duty cycle back through Proposition 3 multiplies the required span by 2.5, equivalently the required ratio by $\sqrt{2.5} = 1.58$. Combined with the corrected floor of 0.39 from Table 5, a variable in this set would need an on-period annualised information ratio of roughly 0.6—more than twice the threshold for economic significance—for its own sample to have established it.

Table 5: The certification floor implied by each variable’s own sample, for the twelve shortest-sampled. “Solo” is a single test at 5%; “corrected” applies Bonferroni over the 46-variable search. Both are compared against an annualised information ratio of 0.25.

Variable	Frequency	Span (years)	Floor, solo	Floor, corrected
rsvix	Monthly	29.6	0.46	0.72
impvar	Monthly	29.6	0.46	0.72
vp	Monthly	31.9	0.44	0.69
vrp	Monthly	34.9	0.42	0.66
crdstd	Quarterly	35.5	0.42	0.66
disag	Monthly	44.0	0.37	0.59
shtint	Monthly	52.9	0.34	0.54
sntm	Monthly	58.3	0.33	0.51
accrul	Annual	60.0	0.32	0.50
cfacc	Annual	60.0	0.32	0.50
csp	Monthly	65.6	0.31	0.48
ndrbl	Monthly	67.8	0.30	0.47
Median over all 56: span 99 y, solo 0.25, corrected 0.39				
Below 0.25: solo 33/56; corrected 0/56				

One degenerate fit, flagged rather than trusted. The variance risk premium at quarterly frequency reports $R_{OS}^2 = -98\%$ alongside an e-value of 4.76. This is not a contradiction: its regression forecasts a -78% quarterly equity premium against an outcome standard deviation of 8%, a nine-sigma artifact of a 120-observation fit. Squared-error measures are quadratic and one such point destroys them; the certificate’s stake is capped relative to the signal’s own scale and is unmoved. We record the largest forecast each fit ever made in units of the outcome’s standard deviation and flag anything beyond five. One of 117 is flagged; excluding it changes nothing.

7 Discussion

A null result and a bound are different objects. Welch and Goyal [2] established that these predictors do not beat the historical mean out of sample, and we reproduce it. What that literature has not supplied is the other half: how large an effect the data could have excluded. Without it a negative result is not interpretable, because it is consistent both with there being nothing and with there being something the design could never have seen. Table 1 shows this literature has lived in the second case throughout.

The instrument is not the finding. We are explicit that the certificate is less powerful than Clark–West where Clark–West is well specified [4]. It is used here for three things a t -statistic cannot do: produce a ceiling, date an edge, and correct a dependent grid without a bootstrap. Its failure to certify is therefore weak evidence on its own, and the ceilings—which do not depend on rejecting anything—carry the paper.

What would change the answer. Not a better test: Proposition 1 binds any procedure. Not finer sampling: Corollary 2. Only a longer span, a larger effect, or a smaller search. Of those, only the last is under a researcher’s control, which is an argument for pre-registering a small number of candidates rather than screening many.

8 Limitations

The floor of Proposition 1 assumes a constant effect; a predictor that is strong in some regimes and absent in others has a higher effective floor still, so our numbers are optimistic. Bonferroni is conservative, and a sharper multiplicity correction would lower the corrected column of Table 1, though not to the point of clearing the boundary. The certificate’s own conservatism means its ceilings are wider than a fixed-sample interval would be, which cuts against us. And our predictor set is Goyal et al.’s: a set assembled from published papers is selected on past significance, which biases toward finding predictability, not away from it.

9 Conclusion

Evidence about an alpha accumulates as log wealth, so what a sample can certify is fixed in advance by its calendar span, its level, and the size of the search. For the canonical equity-premium data those quantities give 99 years for a single predictor against the effect the literature calls economically significant, and 244 once the search that was actually run is accounted for. There are 100 years. Applying an anytime-valid certificate to 117 predictor–frequency pairs certifies nothing under any of nine specifications, but the useful output is the interval it does supply: the median predictor’s incremental annualised information ratio is at most 0.60 and cannot be separated from zero. A century of the best-studied data in finance is not enough to tell the difference between no predictability and twice as much as would matter—and, because evidence compounds, the same instrument shows that the most cited predictors in the literature stopped contributing seventy years ago.

A Every predictor

Table 6 reports all 117 predictor–frequency pairs. “Source” records whether the predictor is the real-time diagonal of a Goyal et al. [1] vintage matrix or the workbook column. A dagger marks the single fit flagged as degenerate—a forecast beyond five outcome standard deviations—which is excluded from the summary statistics in the body and changes none of them.

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Table 6: All predictor–frequency pairs, 1926–2025. Rejecting needs an e-value of 20; the largest observed is 7.47. “Ceiling” is the 95% anytime-valid upper bound on the incremental annualised information ratio.

Predictor	Source	n	R^2_{OS} (%)	Clark–West	e-value	Ceiling
<i>Annual</i>						
ogap	workbook	80	3.66	2.54	4.96	1.48
ndrbl	workbook	57	7.30	2.28	3.14	2.17
tail	workbook	80	−0.28	0.29	2.60	0.49
gpce	real time	68	9.53	2.00	2.59	1.47
ltr	workbook	80	−7.67	1.65	2.13	0.53
pce	workbook	62	10.75	3.22	2.13	1.28
lzrt	workbook	80	−2.36	1.46	2.07	0.91
eqis	workbook	80	0.67	1.59	1.93	1.18
d/p	workbook	80	0.22	0.97	1.83	0.76
lty	workbook	80	−6.44	0.94	1.66	0.99
govik	workbook	68	−3.61	1.04	1.64	1.13
i/k	workbook	68	4.89	2.05	1.58	1.25
tbl	workbook	80	−4.59	0.79	1.55	1.09
skvw	workbook	80	−0.12	0.51	1.54	0.79
fbm	workbook	80	−1.26	−1.04	1.46	0.76
d/y	workbook	80	−0.22	1.35	1.37	0.55
d/e	workbook	80	−2.91	−0.69	1.37	0.62
accrul	real time	50	6.68	1.36	1.32	2.12
b/m	workbook	80	−8.12	0.87	1.27	0.80
dfr	workbook	80	−3.95	−0.18	1.26	0.68
e/p	workbook	80	0.95	1.82	1.26	0.59
tchi	workbook	64	−4.83	−0.68	1.25	0.63
svar	workbook	80	−0.18	−0.23	1.16	0.78
sntm	workbook	48	−15.16	0.06	1.15	1.06
wtexas	workbook	80	−4.34	0.81	1.15	1.22
gip	real time	80	−1.86	−1.11	1.13	0.11
disag	workbook	34	−11.30	0.08	1.11	3.67
skew	workbook	59	−2.91	−0.14	1.10	1.13
cfacc	real time	50	7.69	3.34	1.09	2.66
ntis	workbook	80	−6.87	0.27	1.09	1.06
dtoat	workbook	80	−1.33	−0.34	1.05	0.64
cay	workbook	70	−2.99	0.22	1.03	1.34
house	real time	80	−1.04	0.26	1.01	0.38
rdsp	workbook	80	−0.93	0.33	1.01	0.71
dfy	workbook	80	−0.90	−0.76	1.01	0.61
avgcor	workbook	80	−7.10	−0.71	1.01	0.46
dtoy	workbook	80	−6.60	−0.79	1.01	0.43
ygap	workbook	62	−11.46	0.30	1.00	1.07
shtint	workbook	42	−24.60	−1.58	1.00	2.04
csp	workbook	56	−8.65	−0.97	1.00	0.60
infl	workbook	80	−5.93	−1.53	1.00	0.30
tms	workbook	80	−3.15	0.40	1.00	1.07
<i>Monthly</i>						
wtexas	real time	960	0.01	1.85	7.47	0.35
sntm	workbook	641	0.42	2.03	5.92	0.66
ogap	real time	960	0.26	1.73	5.77	0.46
infl	workbook	960	0.11	0.83	5.71	0.51
ndrbl	real time	743	−0.09	0.99	4.57	0.62
csp	workbook	685	−1.05	2.07	4.30	0.46
tms	workbook	960	−0.08	0.78	3.92	0.78
tbl	workbook	960	0.07	1.44	3.48	0.91
ntis	workbook	960	−0.33	0.86	2.84	0.60
disag	real time	468	−0.66	−0.31	2.11	0.99
vrp	real time	360	−6.20	−0.04	2.04	1.31
lzrt	real time	960	−0.17	0.66	1.93	0.40
lty	workbook	960	−0.60	1.38	1.91	0.54
dtoat	real time	960	−0.07	−0.20	1.70	0.62
d/e	workbook	960	−1.52	−0.37	1.47	0.42
d/p	workbook	960	−0.10	1.34	1.45	0.46
dfy	workbook	960	−0.11	−0.75	1.40	0.38
fbm	workbook	960	−1.57	0.17	1.33	0.42
rdsp	real time	960	−0.06	−0.53	1.26	0.44
tchi	real time	779	−0.07	0.51	1.24	0.70
svar	workbook	960	−0.30	−0.51	1.24	0.41
dtoy	real time	960	−0.55	−0.53	1.23	0.43
rsvix	workbook	296	−3.11	−0.81	1.16	0.89
impvar	real time	296	−1.79	0.14	1.15	1.12
d/y	workbook	960	−0.75	1.64	1.14	0.41
tail	real time	960	−0.37	−0.52	1.08	0.36
skvw	real time	960	−0.61	−1.32	1.04	0.18
avgcor	real time	960	−0.17	−0.18	1.02	0.38
vp	real time	324	0.07	0.63	1.02	1.10
ltr	workbook	960	−0.99	−0.56	1.00	0.27
e/p	workbook	960	−0.79	1.55	1.00	0.40
b/m	workbook	960	−1.35	0.59	1.00	0.46
shtint	real time	515	−9.15	0.99	1.00	1.00
ygap	real time	812	−1.02	−0.46	1.00	0.39
dfr	workbook	960	−0.43	−0.23	1.00	0.36
<i>Quarterly</i>						
i/k	workbook	295	2.94	3.19	5.87	0.65
sntm	workbook	213	1.42	2.28	5.33	0.64

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